SutureAI

Platform User Guide

Recording, scoring and reviewing simulated suturing practice

For students and instructors · Prototype edition · October 2026

What SutureAI does. Students record a practice task (simple suturing, knot tying or needle handling) and upload the video. The platform scores it on five dimensions and writes a report that links each finding to a moment in the video. Instructors follow their whole class, confirm or adjust scores, and decide students' appeals against individual findings.

Students	Section 3: dashboard, uploading a recording, reading a report, appealing a finding, exporting a PDF.
Instructors	Section 4: class overview, student detail, review queue, deciding appeals, reviewing a report.
Everyone	Sections 2 and 5: signing in, statuses, troubleshooting.

1 How it works

Every recording goes through the same loop. The student only does two things (upload, then respond); scoring and the first draft of the report are automatic; the instructor has the final word on the score and decides any appeal against a finding.

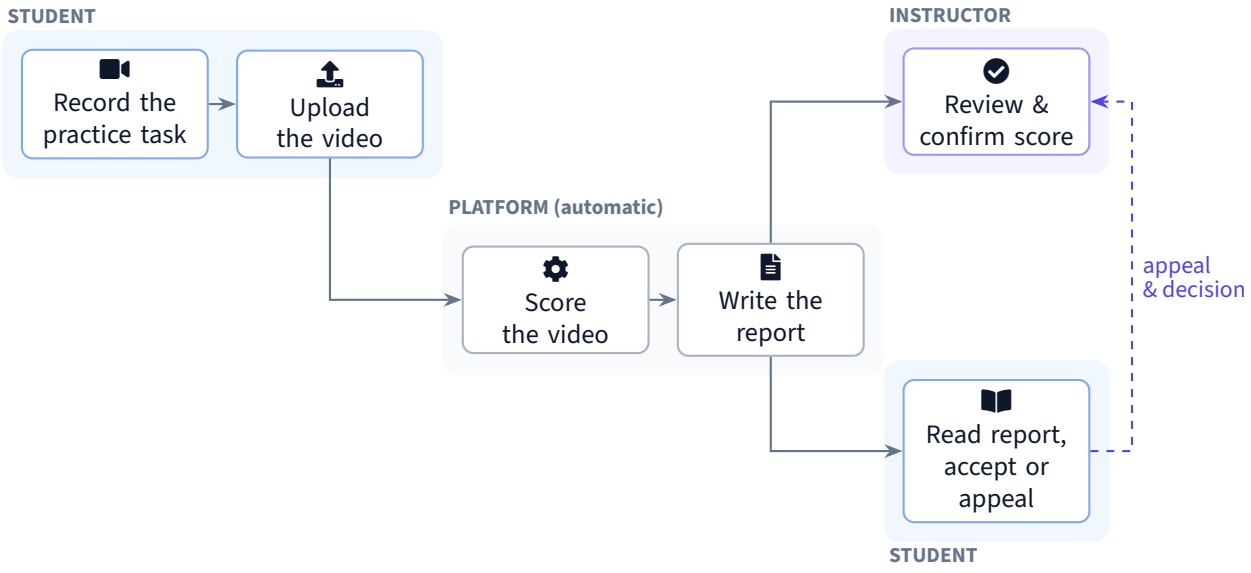


Figure 1: The training loop. Scoring and report writing run in the background; the student can close the page and come back.

The report scores five **dimensions**, each from 0 to 100:

Efficiency	Duration, path length of the instruments, idle time between steps.
Accuracy	Needle entry and exit points, repeated insertion attempts, stitch spacing.
Procedural Flow	Whether every step was done, in the right order, without rework.
Safety	Dropped needle, wrong instrument, instruments leaving the working area.
Product Quality	Consistency of the finished stitches and security of the knots.

Prototype notice. In this version the video scoring step is a placeholder: scores and detected events come from a simulation (or, for the demonstration recording, from a manual annotation) rather than from analysing the video. Everything else (upload, reports, appeals, reviews, PDF export) works as described.

2 Getting started

2.1 Signing in

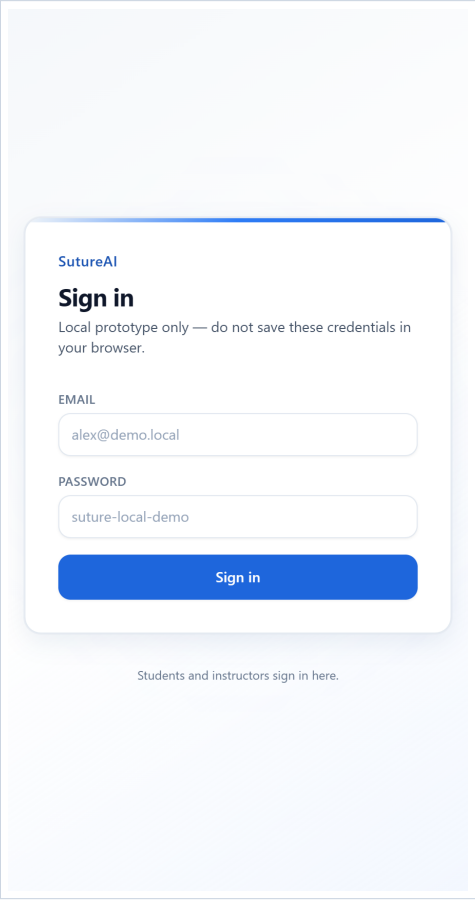


Figure 2: Sign-in page.

Enter your email and password and press **Sign in**. You land on the page for your role: students on their *dashboard*, instructors on the *class overview*.

The demo installation comes with these accounts (password `suture-local-demo`):

Email	Who
`chen@demo.local`	Instructor, Groups A and B
`patel@demo.local`	Instructor, Group B only
`alex@demo.local`	Student, Group A
`priya@demo.local`	Student, scores declining
`sam@demo.local`	Student, improving
`jordan@demo.local`	Student, Group B

Instructors only ever see students in the groups they teach.

3 For students

3.1 Your dashboard

The dashboard (`/`) is your home page. From top to bottom it shows your course and current task, any recording that is still being processed, a snapshot of your latest report, your recent trend, the practice priorities taken from your last report, notifications (for example when your instructor has reviewed a report or decided one of your appeals), a calendar of training days and your full history.

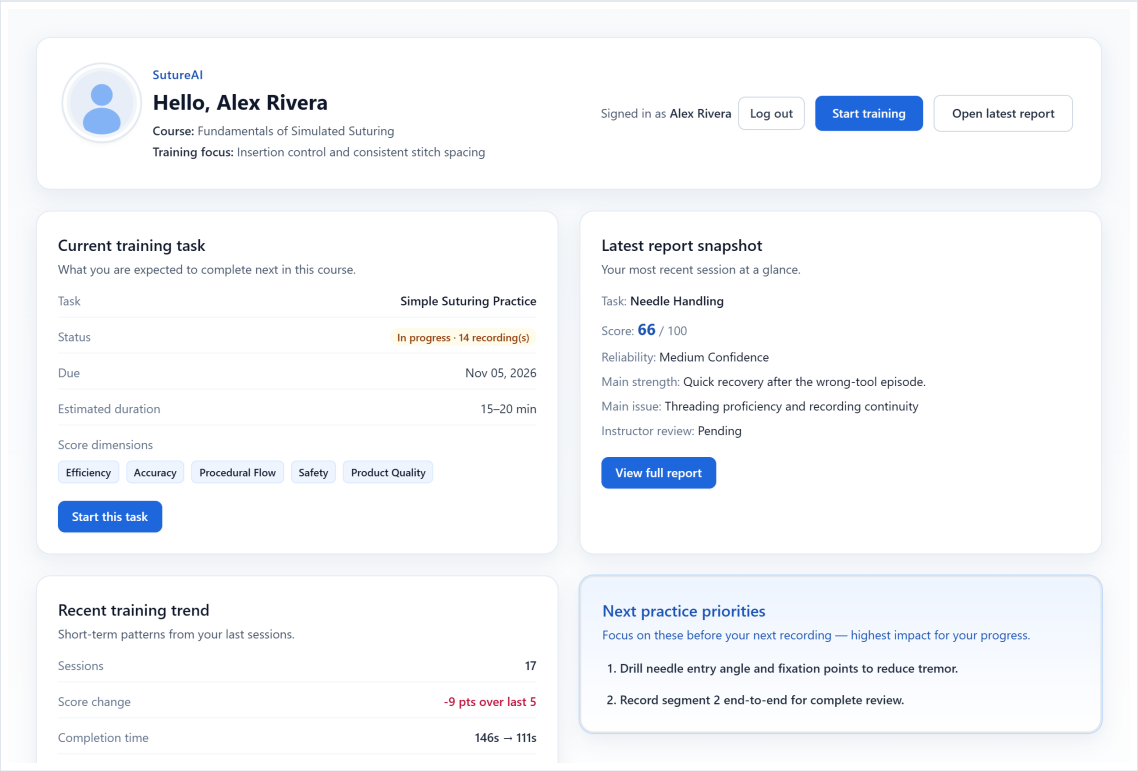


Figure 3: Student dashboard. **Start training** records a new session; **Open latest report** jumps to your most recent report.

3.2 Recording and uploading a session

1. On the dashboard press **Start training** (or **Start this task** on the current-task card).
2. **Pick the task** you practised.
3. **Add your video**: drag it onto the dashed area or click it to browse. MP4, WebM, MOV and MKV are accepted. A preview appears with the file size and duration so you can check it is the right recording.
4. Press **Upload & analyse**. A progress bar shows the upload; you can cancel it.

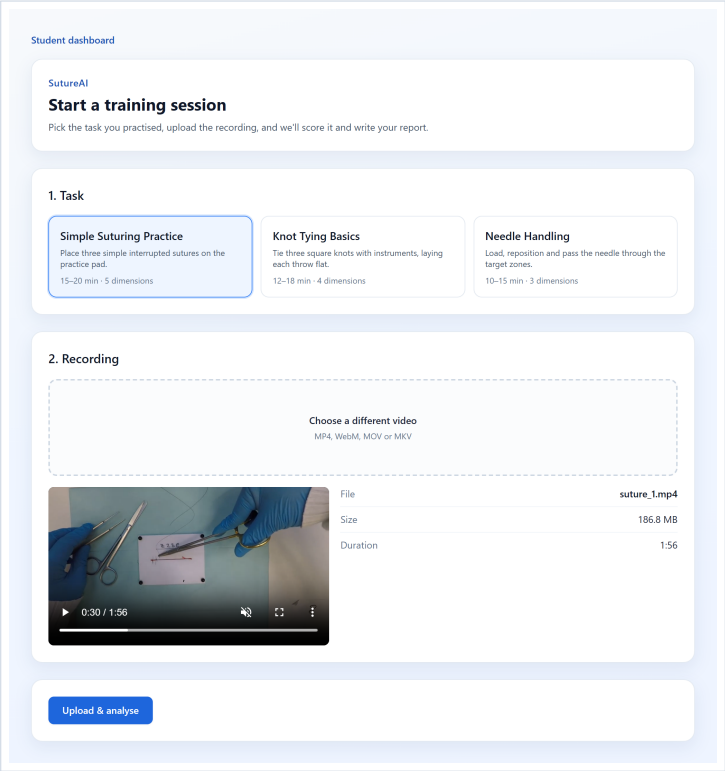


Figure 4: Choosing the task and the video before uploading.

After the upload the page shows the analysis as it happens: the overall percentage with an estimate of the time left, a checklist of the seven analysis stages (verify the recording, detect instruments, track hands, segment the procedure steps, score the dimensions, write the report, link evidence clips) and an *Activity* panel describing what is being done right now. It usually takes under a minute. When it finishes, the report opens automatically.

You can leave the page at any time: unfinished sessions are listed under *Recordings in progress* on your dashboard, and the page address (`/train/<session>`) can be reopened later.

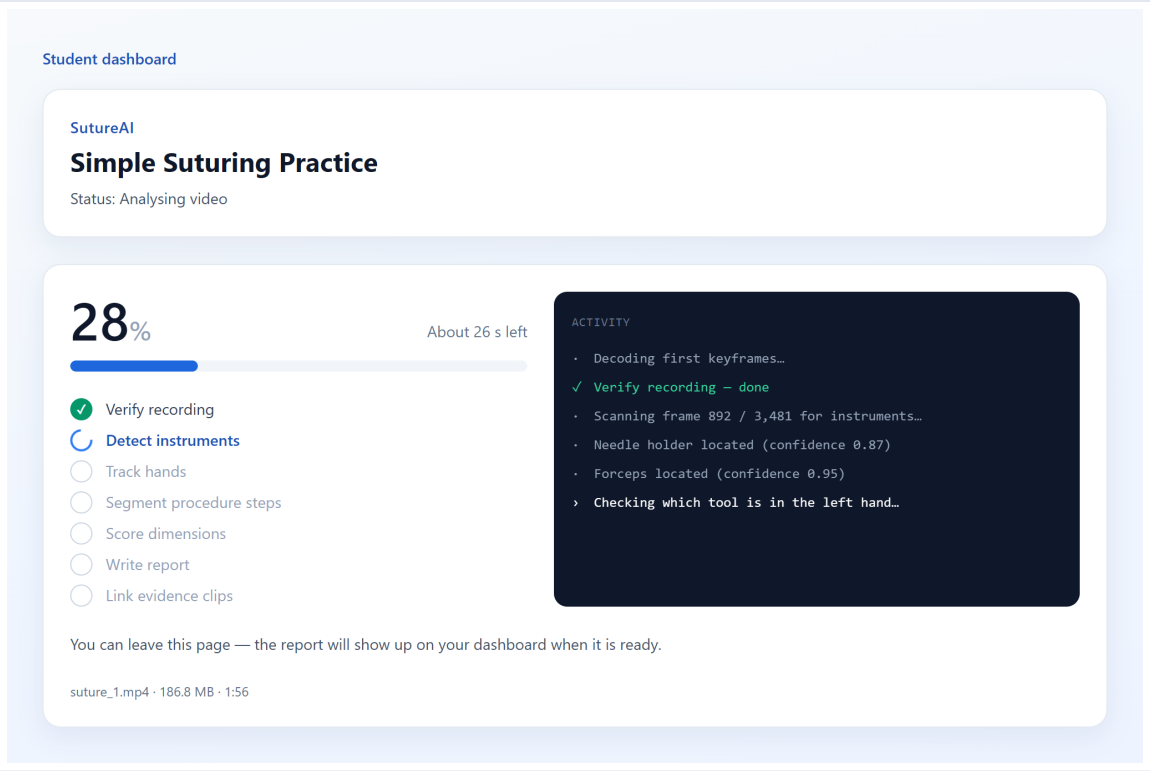


Figure 5: Analysis in progress: stage checklist on the left, live activity on the right.

For the most reliable scores, film from above with the whole practice pad in frame, keep your hands from covering the needle, use even lighting, and record the task from start to finish without stopping.

3.3 Reading your report

The top of the report gives the overall score, how confident the system is, the change since your previous session and the five dimension scores with a radar chart. The badge in the top-right corner shows whether your instructor has reviewed it yet. If your instructor confirmed a different score, or accepted one of your appeals, a *Current score* line shows the score that counts. Use the **Session** menu to switch between your reports.

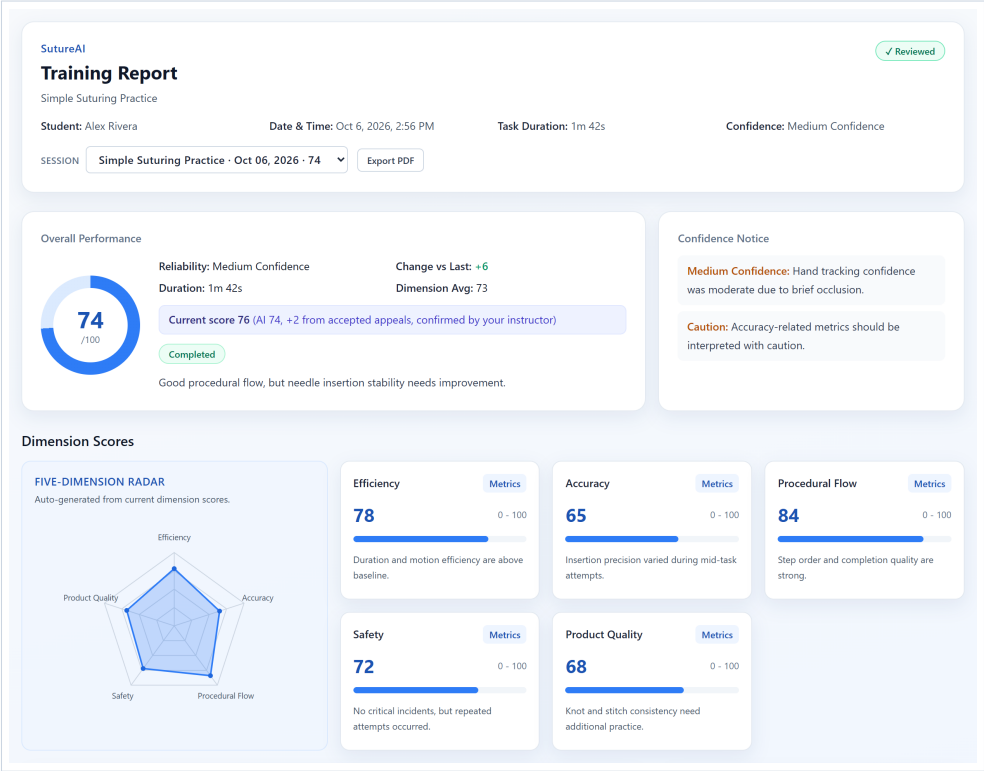


Figure 6: Report header, overall performance and dimension scores.

The **timeline** lists what was detected and when: normal steps (blue), warnings (amber) and issues (red). **Expand** shows the detail; **Replay** (or **Jump**) plays the recording from that moment. Warnings and issues also have an **Appeal** button (Section 3.4). *Assessment highlights* summarise the strongest dimension, the main issue and what to focus on.

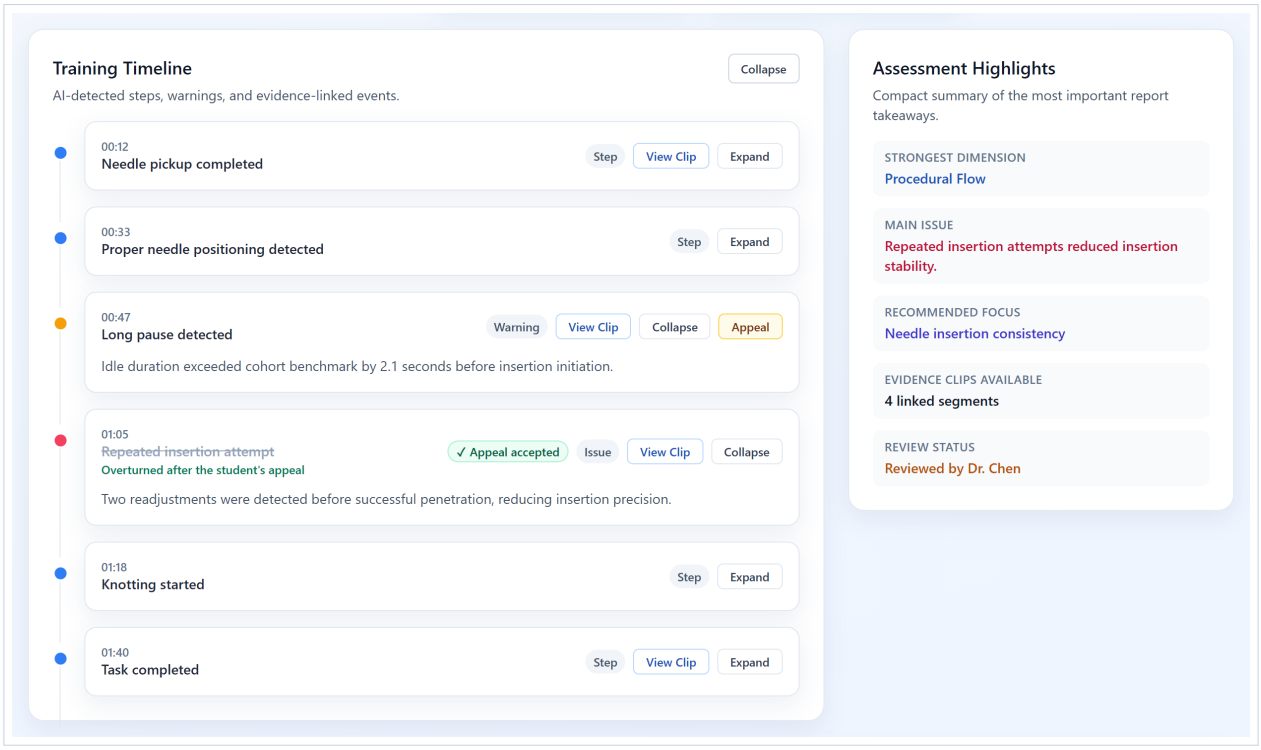


Figure 7: Timeline. The struck-through issue at 01:05 was overturned after the student's appeal; the warning at 00:47 can still be appealed.

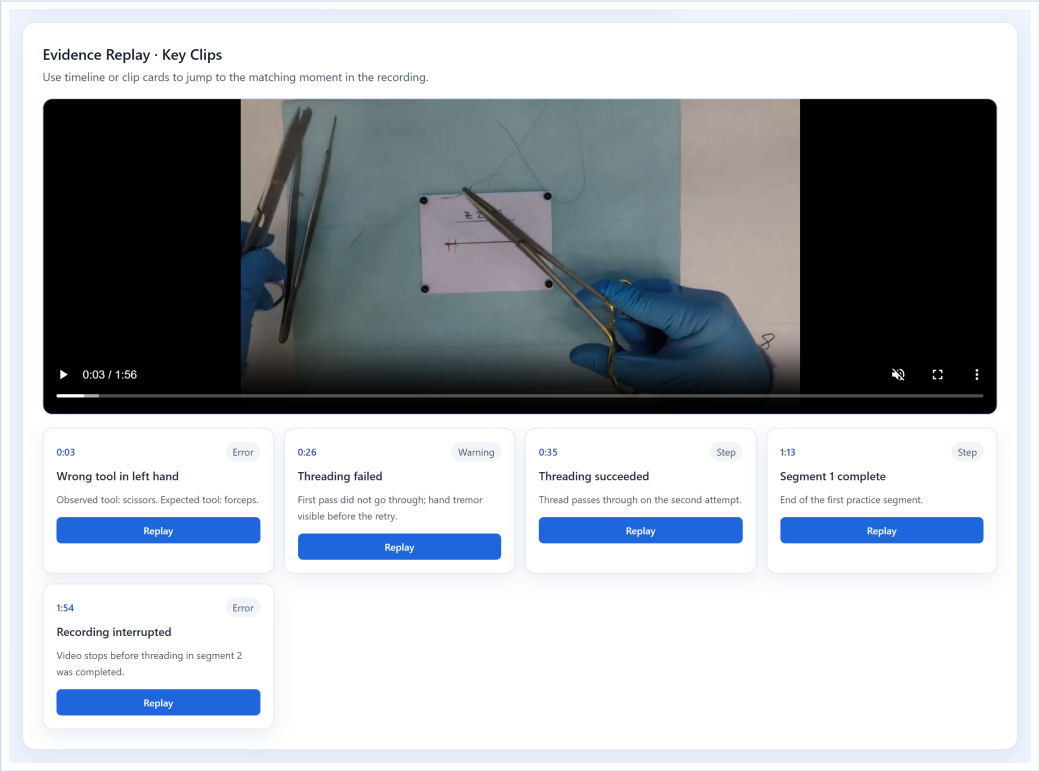


Figure 8: Evidence replay. Each key clip's **Replay** button plays your recording from that moment.

3.4 Accepting or appealing your report

If you agree with the report, press **Accept report** in the *Your response* box at the bottom.

If you think a finding is wrong, **appeal it**:

1. Find the warning or issue in the timeline and press **Replay** to check the moment yourself.
2. Press **Appeal** on that event and choose a reason: *This didn't happen*, *The view was blocked*, *Intentional / wrong moment* or *Other*.
3. Explain what really happened and press **Send appeal**.

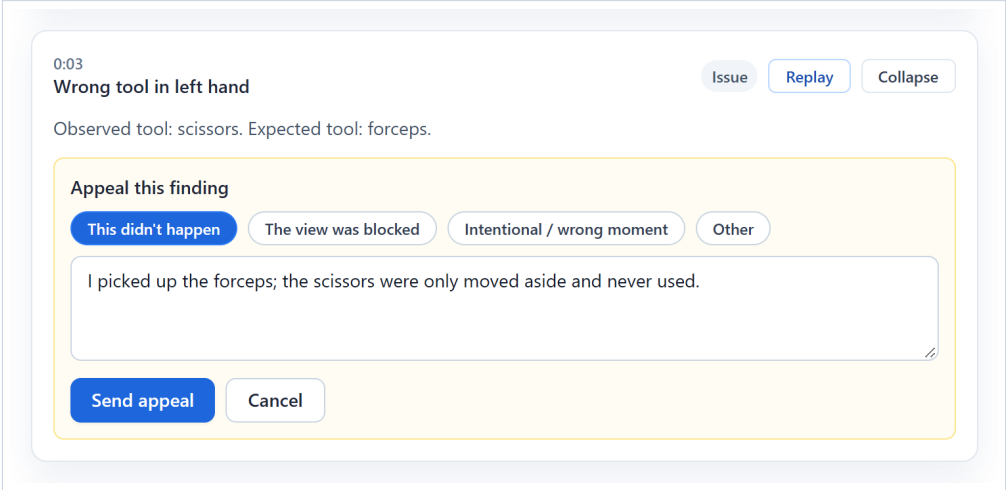


Figure 9: Appealing a finding directly from the timeline.

The event then shows *Appeal pending* and your instructor is notified. They watch the clip and decide:

- **Accepted**: the finding is struck through as *Overturned after appeal* and your score is raised (the badge shows by how much, e.g. *+4*).
- **Rejected**: the finding stays; your instructor explains why.

Either way the decision and the instructor's reasoning appear under *Your response*. You can withdraw an appeal while it is still pending, and you can also appeal the report as a whole with **Appeal the whole report**. A report can carry up to five appeals.

Next to it, **Instructor evaluation** shows your instructor's final score, comment and teaching focus once the review is complete, and **Personal note** is a private reflection saved with the report; instructors cannot see it.

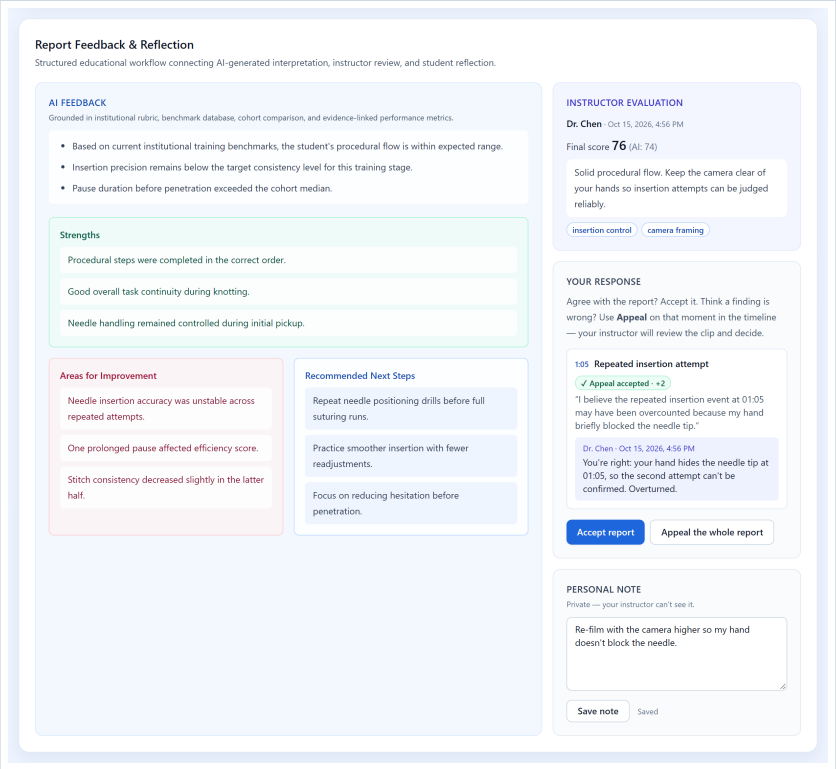
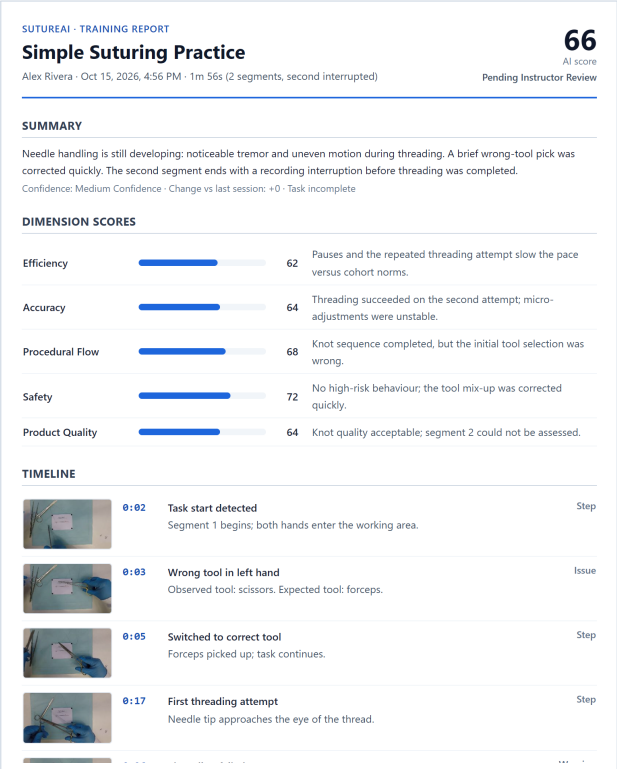


Figure 10: A reviewed report: the appeal on 01:05 was accepted (+2) with the instructor's reasoning, and the student kept a private note.

3.5 Exporting a report as PDF

Press **Export PDF** at the top of a report. A print layout opens in a new tab and the browser's print dialog appears; choose **Save as PDF** as the destination. The PDF contains the scores, the timeline with a still from the recording at every event, strengths and next steps, all appeals with their decisions and the instructor's review, so it can be filed as a record of the session.


SUTUREAI · TRAINING REPORT

Simple Suturing Practice

Alex Rivera · Oct 15, 2026, 4:56 PM · 1m 56s (2 segments, second interrupted)

66
AI score
Pending Instructor Review

SUMMARY

Needle handling is still developing: noticeable tremor and uneven motion during threading. A brief wrong-tool pick was corrected quickly. The second segment ends with a recording interruption before threading was completed.

Confidence: Medium Confidence · Change vs last session: +0 · Task incomplete

DIMENSION SCORES

Efficiency	62	Pauses and the repeated threading attempt slow the pace versus cohort norms.
Accuracy	64	Threading succeeded on the second attempt; micro-adjustments were unstable.
Procedural Flow	68	Knot sequence completed, but the initial tool selection was wrong.
Safety	72	No high-risk behaviour; the tool mix-up was corrected quickly.
Product Quality	64	Knot quality acceptable; segment 2 could not be assessed.

TIMELINE

0:02 Task start detected — Step
Segment 1 begins; both hands enter the working area.

0:03 Wrong tool in left hand — Issue
Observed tool: scissors. Expected tool: forceps.

0:05 Switched to correct tool — Step
Forceps picked up; task continues.

0:17 First threading attempt — Step
Needle tip approaches the eye of the thread.

Figure 11: First page of an exported report. Each timeline event shows the video frame at that moment.

4 For instructors

The instructor area has two pages in the top bar, **Overview** and **Review queue**. Both lead to student pages and reports:

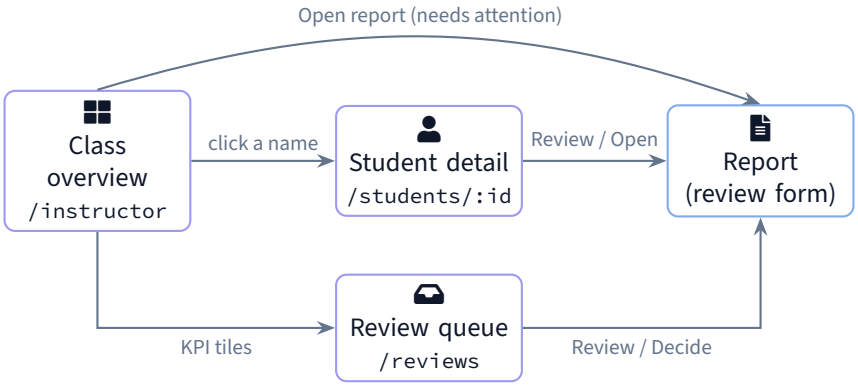


Figure 12: Navigating the instructor area.

4.1 Class overview

Choose a group with the **Cohort** menu; everything on the page refers to that group.

- **Key numbers**: students (and how many trained this week), sessions this week, the 30-day average score, reports waiting for your review and open appeals. The last two are links into the review queue.
- **Needs attention** lists the students to look at now, most urgent first, each with a link to the student or the report.

- **Skill map** shows every student's recent average in each dimension. Paler cells are lower scores, so weak spots (for one student, or for the whole group in a column) stand out.

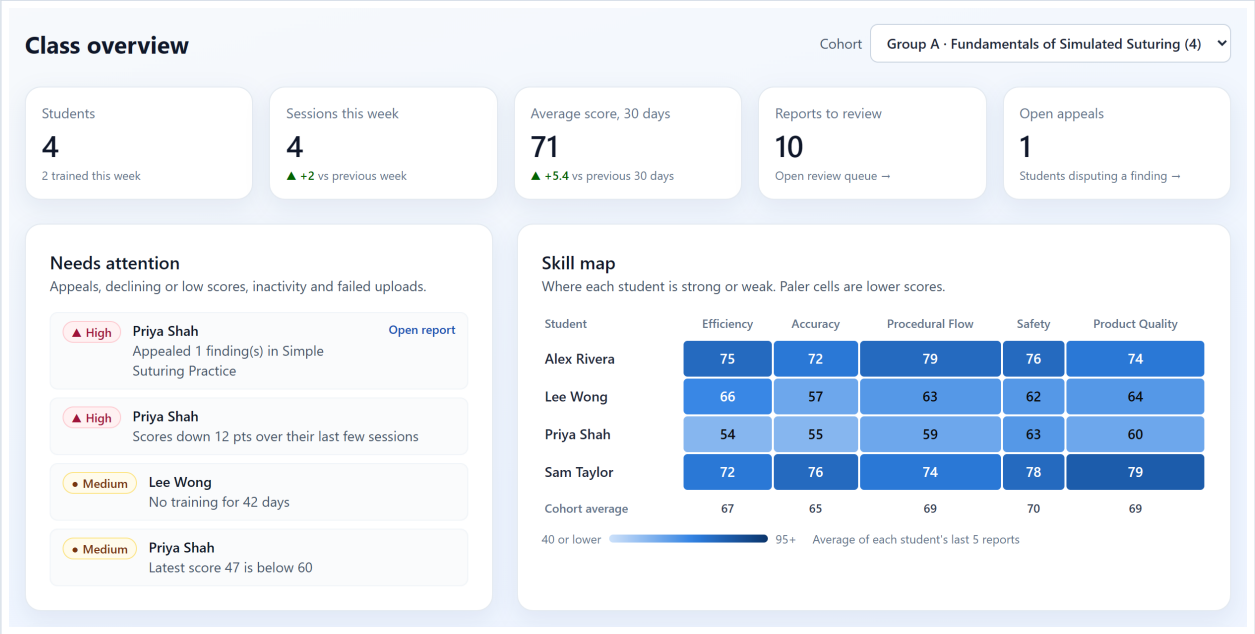


Figure 13: Class overview for Group A.

Flag	Level	When it appears
Open appeal	▲ High	The student appealed a finding and is waiting for your decision.
Declining scores	▲ High	The average of the last 3 reports is at least 8 points below the 3 before them.
Low score	● Medium	The latest score is below 60.
Inactive	● Medium	No training for 14 days or more, or never.
Failed upload	○ Low	A recording could not be processed.

Below the skill map, the **Students** table lists everyone in the group. Click a column heading to sort (by default the students with the most waiting work come first) or type in the search box. *Trend* is a small line of the last eight scores; *Change* compares the average of the last three reports with the three before them.

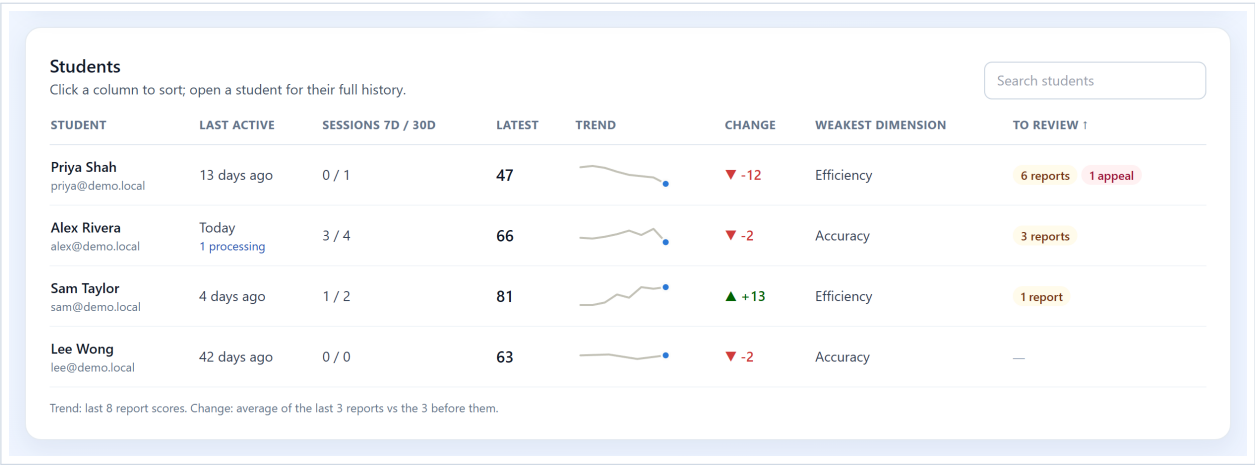


Figure 14: Student roster.

4.2 Student detail

Click a student's name anywhere to open their page: who they are and their current flags, key numbers, the overall score for every report (hover for details, click a point to open that report), one small chart per dimension to see *where* a change comes from, every session with its review and response status, and the training calendar.

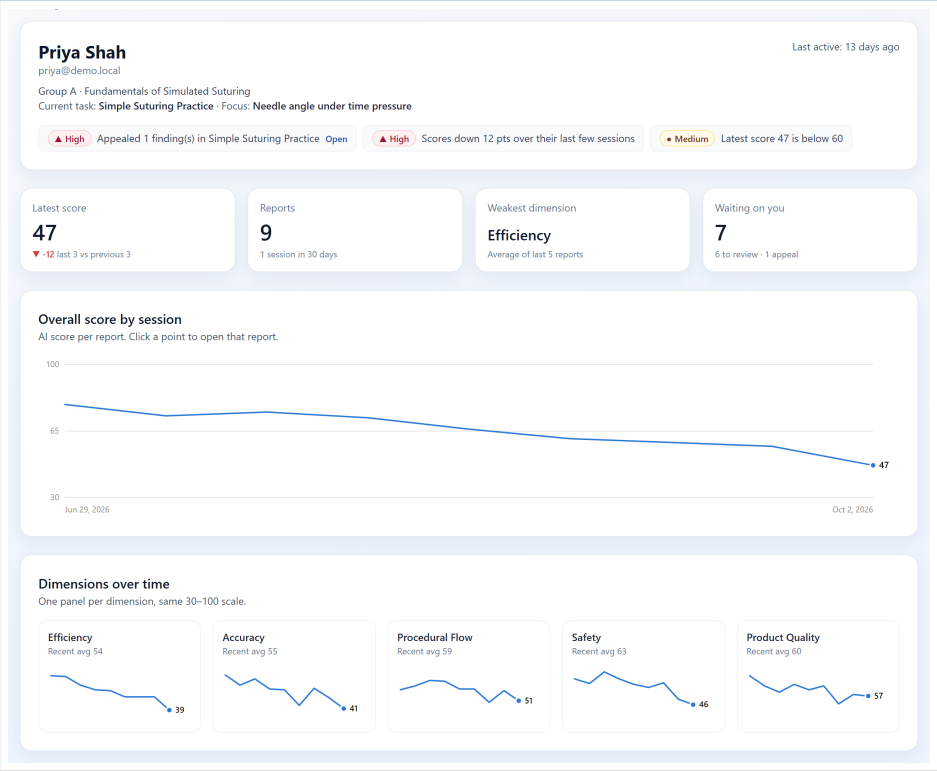


Figure 15: A student whose scores are declining. The dimension charts show the drop is broad rather than in one skill.

4.3 Review queue

The queue collects reports from all your groups (or one, with the **Cohort** filter) in three tabs:

- **To review**: reports without a completed review, oldest first. Drafts stay here until completed.
- **Appeals**: reports with at least one appeal waiting for your decision.
- **Reviewed**: completed reviews, most recent first.

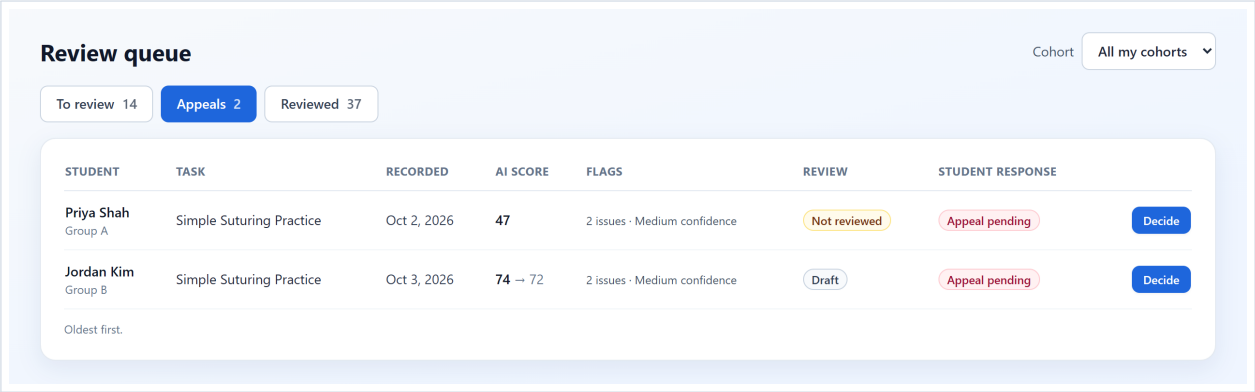


Figure 16: The *Appeals* tab of the review queue. Arrows such as *74 → 72* show a score you changed in a draft.

4.4 Reviewing a report

Opening a report as an instructor shows the *Report Feedback & Review* panel at the bottom.

1. **Decide any appeals** under *Student appeals*. Each one shows the disputed finding, the student's reason and message; ▶ **0:03** replays that moment in the recording. Choose **Accept** (the finding is overturned and the score adjusted by the amount you set; a value is suggested: +4 for an issue, +2 for a warning) or **Reject**, explain your decision, and press **Send decision**.
2. Set the **final score**. It starts at the AI score plus any accepted appeals; the *Alignment* box updates as you type.
3. Write a **comment to the student** and add one or more **teaching focus** tags (suggestions are offered from the weakest dimensions).
4. Press **Save draft** to come back later (the student cannot see drafts), or **Complete review** to send it. A completed review can still be edited with **Update review**.

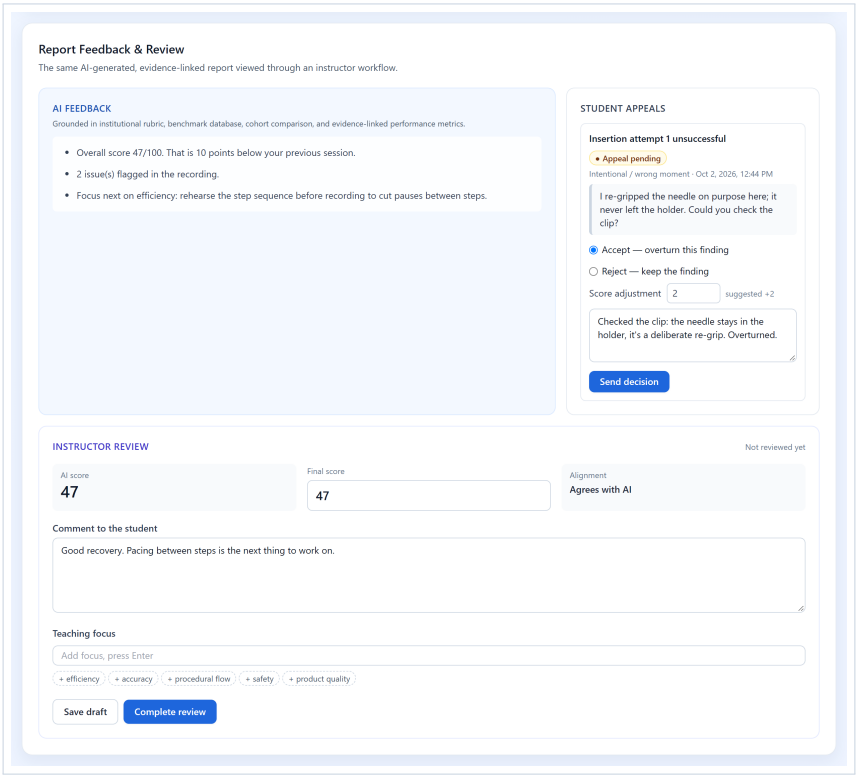


Figure 17: An appeal waiting for a decision, above the review form.

Alignment	Meaning
Agrees with AI	Final score equals the AI score.
Adjusted	Changed by 10 points or less.
Override	Changed by more than 10 points.

If you accept an appeal after completing the review, the final score is adjusted automatically.

The **As student sees it** button at the top of a report previews exactly what the student will see, which is handy before completing a review.

5 Reference

5.1 Session status

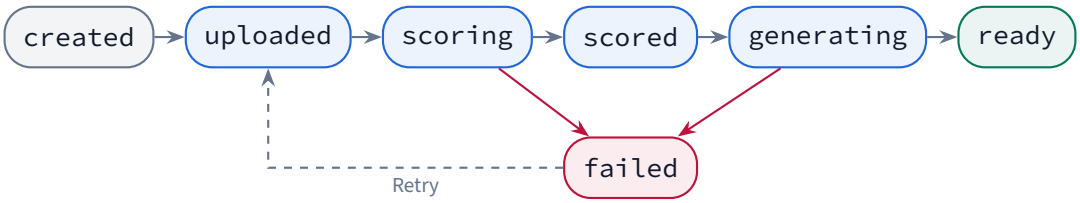


Figure 18: Lifecycle of a training session. *Retry* resumes from the last step that succeeded.

Status	Shown as	Meaning
`created`	Waiting for video	Session started, no video uploaded yet.
`uploaded`	Queued for analysis	Video received, waiting its turn.
`scoring`	Analysing video	The video is being analysed.
`scored`, `generating`	Writing report	Scores are ready; the report is being written.
`ready`	Report ready	The report can be opened.
`failed`	Failed	Something went wrong; the page shows the reason and a **Retry** button.

5.2 Appeal status

Status	Meaning
Appeal pending	Sent; waiting for the instructor. The student can still withdraw it.
Appeal accepted (+*n*)	The finding is overturned and the score raised by *n* points.
Appeal rejected	The finding stands; the instructor's reasoning is shown.

5.3 Troubleshooting

I cannot sign in.
Check the email address and password (the password is case-sensitive). If it still fails, ask your course administrator to check that your account exists.

My video is rejected.
Only MP4, WebM, MOV and MKV files are accepted, up to 1 GB by default. Empty files are refused.

Processing failed.
Open the session (from *Recordings in progress* on the dashboard) and press **Retry**. If it fails again, record a new session with better lighting and framing.

Accept report is greyed out.
You have an appeal pending on that report. Wait for the decision or withdraw the appeal first.

There is no Appeal button on an event.
Only warnings and issues can be appealed, each once, up to five appeals per report.

The print dialog did not open after Export PDF.
Allow pop-ups for the site, or press **Save as PDF** at the top of the print page once the video stills have loaded.

An instructor cannot see a student.
Instructors only see students in the groups they teach.